

Department of Science and Humanities

Course Name:	Basics of Electrical Engineering(BEE)	Semester:	I
Exam:	BEE IA1	Division:	All

Q1. Calculate the current through $30\ \Omega$ resistor using Thevenin's theorem. In the circuit shown in figure 1, the value of R_1 will be the **last two digits of roll no. times Ω**

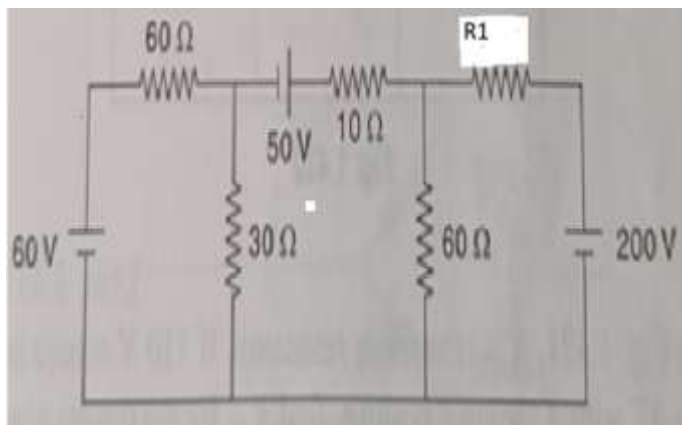
For eg: For Roll no: 1000020, then $R_1 = 20\Omega$. If your last digits are from 01 to 10, kindly add 10 to your last two digits, so roll no with last two digits with 01 will become 11, then $R_1 = 11\Omega$ **(ODD ROLL NO ATTEMPT Q1)**

- Simulate the circuit shown in figure 1, using LTspice software and measure V_{th} , R_{th} and I_L .
- Also solve the numerical theoretically and calculate V_{th} , R_{th} and current through $30\ \Omega$ load resistor (I_L)

Tabulate the results as shown in the table below.

Parameter	Theoretical value	Simulated value
Thevenin's voltage V_{th}		
Thevenin's resistance R_{th}		
Load current I_L		

Figure 1



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Q2. Calculate the current through $4\ \Omega$ resistor using Norton's theorem. In the circuit shown in figure 2, the value of R_1 will be the **last two digits of roll no. time's Ω**

For eg: For Roll no: 1000020, then $R_1 = 20\ \Omega$. If your last digits are from 01 to 10, kindly add 10 to your last two digits, so roll no with last two digits with 01 will become 11, then $R_1 = 11\ \Omega$ **(EVEN ROLL NO ATTEMPT Q2)**

- Simulate the circuit shown in figure 2, using LTspice software and measure V_{th} , R_{th} and I_L .
- Also Solve the numerical theoretical and measure I_N , R_N and current through $4\ \Omega$ load resistor (I_L)

Tabulate the results as shown in the table below.

Parameter	Theoretical value	Simulated value
Norton's current I_N		
Norton's resistance R_N		
Load current I_L		

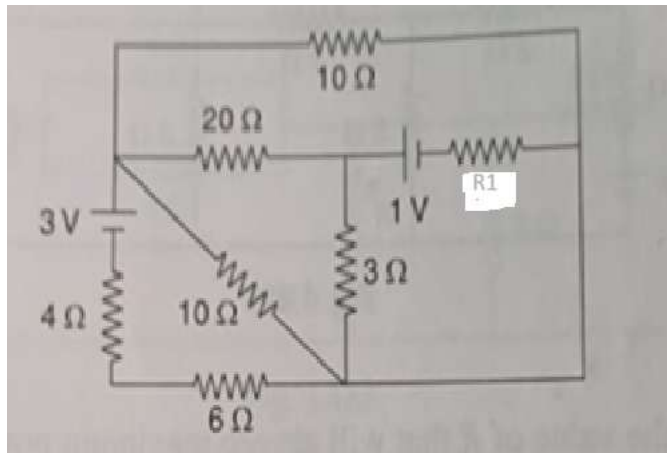


Figure 2